INCH-POUND

MIL-L-24247C(SH)
26 January 1990
SUPERSEDING
MIL-L-24247B(SHIPS)
22 October 1986
(See 6.8)

MILITARY SPECIFICATION

LIFE PRESERVER, VEST TYPE, U.S. NAVY, MARK I

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a combined colored vest and carbon dioxide (CO_2) inflatable life preserver to be worn by Navy personnel for identification and survival purposes. Type I (color 1 through 7) will be worn by flight deck personnel for specific identification. Type II (colors 8 and 9) will be used when authorized.

1.2 Classification. The vest type life preserver shall be of the sizes, colors, and types specified in 1.2.1 through 1.2.3 (see 6.2).

1.2.1 Sizes. Sizing shall be measured in accordance with the following chest measurements:

(a) Small - 34 to 38 inches
(b) Medium - 38 to 44 inches
(c) Large - 44 to 48 inches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, D.C. 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Colors. Colors shall be as follows:

Number		Color
1	White	- Cable no. 70001
2	Old glory red	- Cable no. 70180
3	Irish green	- Cable no. 70168
4	Royal blue	- Cable no. 70087
5	Cocoa brown	- Cable no. 70192
6	Lemon yellow	- Cable no. 70205
7	Purple	- Cable no. 70060
8	Safety orange	- No. 12246
9	Gray	- No. 36081

1.2.3 Types. Types shall be as follows:

Type I - (Colors 1 through 7) Identification and survival vest for flight deck personnel.
Type II - (Colors 8 and 9) Survival vest for authorized use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation. (See 6.2)

SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton.
V-T-295	-	Thread, Nylon.
QQ-B-626	-	Brass, Leaded and Non-Leaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-N-290	-	Nickel Plating (Electrodeposited).
QQ-W-343	-	Wire, Electrical, Copper (Uninsulated).
UU-P-268	-	Paper, Kraft, Wrapping.
DDD-B-1199	-	Binding Textile, Cotton, Bias-Cut.
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage (General Use).

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MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys.
MIL-W-530	-	Webbing, Textile, Cotton, General Purpose, Natural or in Colors.

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MILITARY (Continued)

MIL-B-543 - Buckles, Tongueless and Web Strap.
MIL-C-601 - Cylinders, Carbon Dioxide Filled, Technical.
MIL-W-4088 - Webbing, Textile, Woven Nylon.
MIL-R-5001 - Rubber Cellular Sheet, Molded and Hand Built Shapes; Latex Foam.
MIL-T-5038 - Tape, Textile and Webbing, Textile, Reinforcing, Nylon.
MIL-A-5540 - Adhesive, Polychloroprene.
MIL-R-6855 - Rubber, Synthetic Sheets, Strips, Molded or Extruded Shapes.
MIL-F-10884 - Fasteners, Snap.
MIL-P-12420 - Plastic Material, Cellular, Elastomeric.
MIL-B-17757 - Boxes, Shipping, Fiberboard (Modular Sizes).
MIL-C-19002 - Cloth, Coated, and Strip, Coated Cloth - Polychloroprene on Nylon.
MIL-C-19377 - Cloth, Twill, Nylon. (Colors 8 and 9).
MIL-F-21840 - Fastener Tapes, Hook and Pile, Synthetic.
MIL-I-23145 - Inflation Assemblies, Life Preserver.
MIL-T-43566 - Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors.
MIL-O-81375 - Oral Inflation Assemblies, Survival Equipment, Inflatable.
MIL-C-83489 - Cloth, Coated, Nylon, Polyurethane Coated.

STANDARDS

FEDERAL

FED-STD-191 - Textiles Test Methods.
FED-STD-595 - Colors.
FED-STD-601 - Rubber: Sampling, and Testing.
FED-STD-751 - Stitches, Seams, and Stitchings.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-2482705 - Life Preserver, Vest Type, U.S. Navy, Mk-1.
805-1183395 - Rubber Sleeve Check Valve, for CO_2 Inflatable Life Preserver.

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation. (See 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B 117 - Standard Method of Salt Spray (Fog) Testing.(DoD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COLOR ASSOCIATION OF THE UNITED STATES
Standard Color Card of America

(Application for copies should be addressed to the Color Association of the United States, 200 Madison Avenue, New York, NY 10016.)

SOCIETY OF AUTOMOTIVE ENGINEERS
AMS 3272 - Cloth, Nylon Polyurethane Coated, Thermally Radio Frequency or Ultrasonically Sealable.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Life preservers, vests (covers), or inflatable bladders furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.5).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Material. In instances where the drawings referenced herein do not agree with the materials specified in this specification, the drawing shall govern.

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3.3.1 Vest.

3.3.1.1 Description. The preserver vest shall be cardigan style, sleeveless, with snap fastener front closure, side webbing, with buckle adjustments, and shall be fully lined. The right front shell shall have an oral tube keeper, two reinforced openings, an inflation assembly strap, a vertical slit with a snap fastened cover extending upwards from the lower reinforced opening, and a CO_2 cover with a whistle pocket. The left front shell shall have a distress light marker pouch, and a sea dye marker pouch.

3.3.1.2 Cotton. The outer shell and lining for colors 1 through 7 shall be of cotton cloth made from combed cotton having a warp of 40/2 ply and a filling of 15/1. The cloth shall be preshrunk and treated with a durable water repellant finish. The cloth shall weigh 6.0 ± 0.3 ounces/square yard and have a minimum warp of 101 yarns/inch and a minimum filling of 44 yarns/inch. The cloth shall have a minimum breaking strength of 140 pounds warp and 55 pounds filling. The minimum tear strengths shall be 7 pounds for both warp and filling. Maximum allowable shrinkage shall be 2 percent for both warp and filling. Test methods for the above physical requirements shall be in accordance with FED-STD-191. The color shall be as specified (see 6.2.).

3.3.1.2.1 Nylon. The outer shell and lining for colors 8 and 9 vests shall be of nylon cloth conforming to MIL-C-19377. The color shall be as specified (see 6.2.).

3.3.1.3 Thread.

3.3.1.3.1 Cotton. The cotton thread used for colors 1 through 7 shall conform to type IA3, 4-ply, ticket number 16 of V-T-276. The color shall be black and shall be produced by vat dyes.

3.3.1.3.2 Nylon. The nylon thread used on the inflatable bladder and on the covers for colors 8 and 9 shall conform to type I, class A or B, size E of V-T-295. The color shall be black.

3.3.1.4 Binding.

3.3.1.4.1 Colors 1 through 7. The bias-cut binding for the edges of the vest and all slits shall conform to type II, class 2, 1 to 1-1/2 inches wide of DDD-B-1199. The color shall be olive green.

3.3.1.4.2 Colors 8 and 9. The binding for the edges of the vest and all slits shall conform to type III, 1-inch wide of MIL-T-5038. The color shall be olive green.

3.3.1.5 Fasteners. Snap fasteners shall conform to style 2, finish 2 of MIL-F-10884, except front snap fasteners shall conform to style 4, finish 2.

3.3.1.6 Buckles. Buckles for the waist adjustment (take-up) straps shall conform to type II, class 1, style 4, construction A or B, size 3/4-inch of MIL-B-543. The material shall be brass equal to or better than the requirements of QQ-B-626 and shall have a lusterless black finish conforming to MIL-F-495.

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3.3.1.7 Webbing and tape.

3.3.1.7.1 Webbing. The webbing for the waist adjustment (take-up) straps shall conform to type Ia, class 2, 3/4-inch wide of MIL-W-4088. The color shall be olive drab. The webbing for the whistle pocket tab and the distress light tab shall conform to Type II class II of MIL-W-4088, the color shall be Air Force yellow 1365.

3.3.1.7.2 Tape. The tape used for the 4-1/2 inch slit covering shall conform to type I, 1-inch wide, of MIL-T-43566. The color shall be olive drab.

3.3.1.8 Chafing pad. The chafing pad used in the neck area of the bladder shall be constructed as shown on Drawing 803-2482705. The pad shall be of sponge rubber conforming to type II, grade B, class medium of MIL-R-5001. It shall be backed with fabric as specified in MIL-R-5001. All parts shall be sewn together as a single unit and shall then be cemented, heat sealed, or sewn as shown on Drawing 803-2482705.

3.3.1.9 Tape fasteners. Nylon tape fasteners used to secure the bladder to the cover shall conform to type II of MIL-F-21840. The hook portion of the fastener shall be cemented to the bladder. The pile portion shall be attached to the cover in such a manner as to mate with the hook portion as shown on Drawing 803-2482705.

3.3.1.10 Protective cover flap for CO_2 inflation system. The protective cover flap shall be the same color and material as the outer shell.

3.3.1.11 Reinforcement material. The front edge, and snap fasteners, shall be reinforced with coated cloth conforming to type I of MIL-C-19002.

3.3.1.12 Reflective tape. The material for the reflective tape used on the shoulders of colors 1 through 7 shall be bright silver, and conform to CFR 46 Part 164.018, Type I.

3.3.2 Inflatable bladder.

3.3.2.1 Coated cloth. The coated cloth used in construction of the inflatable bladder shall be heat-sealable polyurethane coated nylon conforming to AMS 3272 Type I except that tear strengths shall be 3.75 pounds for warp, 2.75 pounds for fill, and 45 pounds for adhesion.

3.3.2.1.1 Color of coated cloth. The coated cloth color shall be in accordance with FED-STD-595. Unless otherwise specified herein, the choice of color shall depend on material availability.

3.3.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be

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interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Construction. The vest type preserver shall be constructed in accordance with the operations shown in table I. The contractor is not required to follow the exact sequence of operations provided that the finished vests are identical to those produced by following the sequence as listed in table I.

TABLE I. Construction.

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
Cutting. The vest shall be cut as follows: (a) The vest shall be cut in strict accordance with the drawings furnished, which will show size, shape, and directional lines for cutting and dimensions for proper assembly of parts. (b) Both the left and right front facings shall be cut from the selvage edges of the material Replacement of damaged parts. Damaged parts shall be replaced as follows: (a) During the spreading and cutting operations, any material found to contain defects which may seriously affect appearance or serviceability of the end item shall be excluded and replaced with properly matched material. (b) Any defective part which may seriously affect appearance or serviceability of the end item shall be excluded and replaced with properly matched materials. (c) Any part damaged during the construction process which may affect appearance or serviceability of the end item shall be excluded and replace with properly matched material.					

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
Right front. Oral tube keeper (vertical position). The keeper shall be constructed as follows:					
(a) Cut a piece of 5-1/2 ± 1/4 inches wide cotton in accordance with section 3.3.1.2 to a length of 4-1/4 ± 1/4 inches. Color should be the same as the cover.					
(b) Fold right sides together lengthwise and stitch both sides of horizontal dimension together. Turn inside out so rough edges are inside. Fold cut long edges inward 3/8" and stitch both sides of long dimension (Placed vertically on front) to front, in accordance with drawing, bartacking each end 3/8-inch.	301	LSa-1 and SSc-1	8-12	00/3	00/3
	BRTCK		21 PER BRTCK	00/3	00/3
(c) The keeper shall have 1/2 to 3/4-inch slack (fullness) between the two vertical lines of stitching.					
CO_2 Holder Strap (horizontal position). The CO_2 holder strap shall be constructed as follows:					
(a) Cut a piece of webbing in accordance with MIL-W-4088, Type II, Class 2 to a length of 4-1/4 ± 1/4 inches. Sear the cut ends. Color shall be olive drab green.					

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
(b) Box stitch strap to cover in accordance with the drawing Each box shall be 5/8" by 5/8" and diagonally stitched from corner to corner.	301	LSa-1 and SSc-1	8-12	00/3	00/3
(c) The inflation assembly strap shall have 1/2 to 3/4-inch slack (fullness) between the two boxes.					
Reinforcement Patches. The reinforcement patches for the manifold hole and oral tube hole are attached as follows:					
(a) Stitch inner circle edge of both patches and cover face-to-face on right front as on drawing.	301	SSe-2(a)	8-12	00/3	00/3
(b) Turn patches to finished position (inside the cover) and top stitch inner edges.	301	SSe-2(b)	8-12	00/3	00/3
(c) The circular openings shall be 1/2 to 5/8 inch in diameter.					
(d) Stitch outer edges of circular patches to front, in accordance with drawing, 1/8-inch gauge.	301	LSa-1	8-12	00/3	00/3
(e) Position reinforcement patches according to placement shown on drawing. Attach webbing to outer shell and reinforcement patch using a 7/8 inch square boxstitch through all layers. (PC 97)	301		8-12	00/3	00/3
CO_2 assembly pass-through slit. A vertical slit 4-1/2 inches long shall be formed in the right front outer shell so that the bottom of the slit terminates in the opening for the CO_2 system. The slit shall bisect the opening and shall be bound with binding in accordance with 3.3.1.4. (PC 54)	301	BSa-1	8-12	00/3	00/3

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
Cover for slit opening. The cover for slit opening shall be constructed as follows:					
(a) A piece of tape in accordance with MIL-T-43566, Type I, color olive drab green approximately 4-1/4 inches long by 1 inch wide shall be placed under the front edge of the shell slit, so that the lower edge of the tape is tangent to the top of the circular opening. Join tape to edge with double needle seam, so that open side faces back of vent.	301	LSc-2	8-12	00/3	00/3
(b) There shall be two 5/8-inch snap fasteners on the tape placed as follows: Place one socket snap fastener at each outer corner of cover. Place two stud snap fasteners reinforced with material, type I of MIL-C-19002 on each front to correspond with the socket snap fasteners on the cover.					
Cover for CO_2 system and CO_2 holder strap. The cover shall be constructed as follows:					
(a) Stitch cover on 3/8" from edge. Position whistle pocket tab as shown on drawing.	301	SSa-1	8-12	00/3	00/3
(b) Turn CO_2 cover to finished position and insert foam interlining to fit snugly inside cover.					
(c) Fold the four raw edges of the whistle pocket cover 1/4 inch under and stitch 3/16 inch from each edge.	301	LSd-1	8-12	00/3	00/3

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
(d) Place the whistle pocket cover horizontally on the inner side of the CO_2 cover assembly 1 inch below top edge. Position the whistle pocket covers slanted edge even with the slanted edge of the CO_2 cover. Attach to CO_2 cover by stitching each long dimension side 1/8" from the edge to form a 1-5/8" opening as shown on drawing. Then, stitch a pleat closed end 1/8" from edge.	301	SSa-1	8-12	00/3	00/3
(e) Fold opposite end of whistle pocket tab 1 inch under to form a loop and bartack across 1/4 inch from end.	BRTCK		21 PER INCH	00/3	00/3
(f) On the nylon webbing outside surface, secure a socket snap fastener on center 1-1/4 inch from the slanted edge of the CO_2 cover. On the whistle cover, secure a stud fastener, 1/2 inch from the top edge.					
(g) Position CO_2 cover over lower reinforcement patch, inflation assembly strap, and CO_2 assembly pass-through slit in accordance with drawing. Turn in top raw edge of cover 1/4" and topstitch to front, 1/8" from edge across width at top, and along the straight side for its entire length, bartacking each end.	301	LSa-1 and SSc-1	8-12	00/3	00/3
	BRTCK		21 PER INCH	00/3	00/3
(h) The CO_2 cover shall finish 5 ± 1/4 inches wide at top, 6 ± 1/4 inches wide at bottom and 8-3/8± 1/4 inches long along straight edge.					

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
Snap fasteners on CO_2 cover. There shall be two 5/8-inch snap fasteners on the CO_2 cover, placed as follows: (a) One socket snap fastener 3/4 of an inch above bottom edge of slanted side of CO_2 cover. (See Detail 18-C) (b) One socket snap fastener on slanted side 4-1/4 ± 1/4 inches from corner snap fastener. (c) Place two stud snap fasteners reinforced with material, type I of MIL-C-19002, on left outer shell to correspond with stud snap fasteners on CO_2 cover. Fasteners shall be positioned so that there is a 1/2- to 3/4-inch slack in the CO_2 cover when fastened.					
Grommet on CO_2 cover. There shall be a grommet in accordance with QQ-B-626 located on the CO_2 cover as shown on detail "18-C". (Finish - Black)					
Left front.					
Distress light pouch. The pouch shall be constructed in accordance with Drawing 803-2482705 and as follows:					
(a) Overlap notches on pouch cover and stitch each notch. (PC 77)	301	SSa-1	8-12	00/3	00/3
(b) Stitch curved portion of pouch bottom to pouch cover botton 1/4" from edge. (PC 91)	301	SSa-1	8-12	00/3	00/3
(c) Double stitch bias binding tape to the straight length of the reinforcement backing. (PC 54)	301	BSc-1	8-12	00/3	00/3

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
(d) Position reinforcement backing along top of pouch cover aligning the curved portions. (PC 90)					
(e) Bind outer edges of pouch cover, reinforcement backing, and pouch bottom straight edge with bias binding tape with single stitch 3/16" from edge. (PC 54)	301	BSc-1	8-12	00/3	00/3
(f) Place two grommets conforming to QQ-B-626 on pouch cover as located on drawing. (PC 79)					
(g) Webbing shall conform to MIL-W-4088, type II, class 2. Sear the cut ends. Place the pull webbing on center of pouch cover, 1/2 inch above the grommets, and secure with button and socket (PC 35 and 36) to the pouch cover and reinforcement backing. Place the fastener stud and eyelet on outer shell left side (PC 32 and 34) to correspond with pouch cover fastener.	301	LSa-1	8-12	00/3	00/3
(h) Locate and align the pouch assembly as shown on drawing and attach to outer shell left side by double stitching the straight edges along the bias binding and bartack both ends as shown on detail "22-A".	301	LSb-2	8-12	00/3	00/3
Sea dye marker pouch. The pouch shall be constructed in accordance with Drawing 803-2482705 and as follows:					
(a) Bind outer edge of bottom sea dye marker pouch with bias binding. Stitch over binding full width of pocket 3/16" from edge.	301	BSc-1	8-12	00/3	00/3

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
(b) There shall be a 5/8-inch stud snap fastener placed in center of pocket width and 2 inches from pocket bottom fold line. There shall be a corresponding socket snap fastener on pocket flap. (c) Fold bottom of piece upwards along lower fold line as shown on drawing. Bind outside of pouch piece with bias binding tape. The folded edge shall not have bias binding tape. (d) Place pouch assembly on left outer shell so lower edge is 1 inch from bottom edge of left outer shell and 2-3/4 inches from outside edge of front binding, as shown on the drawing. (e) There shall be a grommet in accordance with QQ-B-626 located on pouch 1 inch from horizontal edge as shown on drawing. (See Detail "21-E").	301	BSc-1	8-12	00/3	00/3
(f) Double top stitch pouch to left outer shell front along assembly bottom and on both sides. Stitch pocket to outer shell 3/8" below fold line as shown on drawing. Do not attach the flap to shell.	301	SSa-2	8-12	00/3	00/3
Front Reinforcement Tape. The interlining reinforcement conforming to MIL-C-19002, Type I, shall be stitched to the inside of each front edge, 1/8-inch gauge.	301	SSa-1	8-12	00/3	00/3
Shoulders. The front shall be joined to the back at the shoulders with double needle seams. (45, 46, or 47)	301	LSc-2	8-12	00/3	00/3

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
Reflective tape. Reflective tape shall be attached in accordance with the drawing in both shoulder and back areas. Tape shall be attached by stitching 1/8 ± 1/16 inch around perimeter. Tape areas on shoulders shall be further stitched diagonally from corner to corner. Cotton thread conforming to type IA3, 4-ply, ticket number 16 of V-T-276, or equivalent shall be used for stitching.					
Sides. Join the front to back at sides with double needle seams.	301	LSc-2	8-12	00/3	00/3
Lining front. The lining front shall be made as follows:					
(a) Bind horizontal opening at bottom of each front and at center of back with bias binding of sufficient length so that no raw edges will be visible when the unit is completed. (See Reference Dwg.)	301	BSa-1	8-12	00/3	00/3
(b) The openings shall be 7-1/2 ± 1/2 inches long.					
Lining shoulders and sides. Join the front to the back at shoulders and sides with double needle seams.	301	LSc-2	8-12	00/3	00/3
Label. The label shall be affixed as follows: Stitch the identification label on all four sides, 1/8-inch gauge, on the inside center back of vest, with the top of the label 1-1/4 ± 1/4 inches below edge of neck binding.	301	LSa-1	8-12	00/3	00/3

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
Assembly. The assembly shall be constructed as follows:					
(a) Stitch outer shell to lining, back to back, along outer edge at front, bottom, neckline, and armholes, matching sides and shoulder seams. Use a 1/16 to 3/16 inch closing seam.	301	SSa-1	8-12	00/3	00/3
(b) Stitch front to lining, at sides and across bottom of lining opening, in accordance with referenced drawings. Stitching shall extend from armhole vertically to horizontal lining opening and across bottom edge of stitching on lining opening.	301	LSa-1	8-12	00/3	00/3
(c) Stitch back to lining from armhole to armhole across back along bottom edge of binding on lining opening, in accordance with drawings.	301	LSa-1	8-12	00/3	00/3
(d) Bind outer edge of vest and armholes with bias binding having a finished binding width of 3/8 inch and a binding seam from the outside edge to be 1/4 to 5/16 inch.	301	BSc-1	8-12	00/3	00/3
Front snap fasteners. There shall be four 5/8-inch snap fasteners, style 4, finish 2, placed 5/8 + 1/8 inch from right front edge, oriented so that single point pull is at top of fastener, measured from center to center of fastener starting with one stud snap fastener at top, the remaining three as shown on Drawing 803-2482705. Four 5/8-inch socket snap fasteners shall be placed on left front edge, 5/8 ± 1/8 inch from edge, to correspond with stud snap fasteners on right.					

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TABLE I. Construction. - Continued

Description of operation	Stitch type (see 3.4.2)	Seam and stitching type (see 3.4.2)	Stitches per inch	Thread	
				Needle	Bobbin or Looper
Side adjustment. Each side adjustment shall consist of a buckle and two pieces of webbing. Front webbing. The front webbing shall be attached as follows: (a) Box and cross stitch one end of webbing to side, bottom, front, as in drawing, stitching through shell and lining. (b) Thread other end of webbing through first bar (closest to tongue) and fold end under 1 inch, and box stitch across width and both edges of webbing. (c) Free end of webbing shall finish 4-1/2 ± 1/4 inches long. Back webbing. The back webbing shall be attached as follows: (a) Thread webbing through last bar on buckle and with raw edges butted together, box and cross stitch across width and both edges of webbing to side, bottom, back in accordance with markings on drawing, stitching through shell and lining. (b) The free end of webbing shall finish 1 ± 1/8 inch long. Cleaning. The life preserver shall be cleaned as follows: (a) Remove all spots and stains. (b) All ends of stitching shall be trimmed and loose threads removed throughout the vest. (c) Close snap fasteners.					

3.4.1 Sizes and measurements. Unless otherwise specified herein, all measurements and tolerances of the finished vest shall be expressed in inches and shall be in accordance with the measurements shown in table II.

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TABLE II. Sizes and measurements.

Attributes[1]	Small (inches)	Medium (inches)	Large (inches)	Tolerance (inches)
(a) Chest	43	48	53	± 1/2
(b) Length	26-3/4	27-3/4	28-3/4	± 1/2

[1]All measurements shall be taken with the vest front fastened and laid flat.

(a) Chest. Double the measurement taken at bottom of armhole across front of chest to opposite armhole.
(b) Length. Measurement shall be taken at top center of back neck section to the bottom edge.

3.4.2 Stitches, seams, and stitching. Stitch, seam, and stitching types, as specified in table I, shall conform to FED-STD-751. Whenever two or more methods for seams and stitching are given for the same part of the operation, either may be used. Double stitch and double lapped seams shall be 1/4- to 5/16-inch gauge with the outer row of stitching 1/16 inch from the finished edge.

3.4.2.1 Stitches per inch. The range of stitches specified in table I shall be the minimum and the maximum number permitted.

3.4.2.2 Tacking and backstitching. The ends of seams and stitching shall be tacked or backstitched not less than 3/8-inch. Thread breaks or skipped stitches on 301 stitch type of FED-STD-751 shall be secured by stitching back of the break not less than 1/2 inch.

3.4.3 Bladder construction. All air retaining seams in the bladder shall be constructed by heat sealing. Unless otherwise specified herein, in attachment of all other parts to the bladder, either cementing or heat sealing shall be used. In all cementing operations, the surfaces to which the adhesive is to be applied shall be thoroughly washed, if necessary, with a solvent in such a manner that the dusting materials (zinc stearate or talc) or any other surface contaminant is completely removed. The surface shall be cleaned prior to cementing. Neither the coating nor the base undercloth shall be damaged, nor the adhesion between both impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive and shall leave no residue. Cemented areas shall be rolled and stitched out and all trapped air and wrinkles that would impair the function of the preserver shall be removed. Adhesives shall be controlled to ensure that old adhesive is not used. When a two-part adhesive is used, a fresh batch shall be mixed at least every 8 hours during the course of manufacture. Containers for adhesive shall be free from congealed adhesive before being refilled.

3.4.3.1 Cemented area. The adhesive on all outside cemented areas and patches shall be applied to the area and shall extend not less than 1/8 inch and not more than 3/8 inch beyond the edges of the patch and shall be parallel thereto.

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3.4.3.2 Sealed or sealed and cemented areas. The construction of heat sealed seams and the attachment of all parts to the bladder shall be undertaken utilizing the techniques and precautions specified in 3.4.3, so that prior to submission of the completed preserver for final approval, all such seams and attachments shall develop optimum properties of adhesion specified in MIL-A-5540 for cemented areas and AMS-3272 Type I for heat sealed areas.

3.4.4 Oral inflation. The oral inflation valve (see 3.5.3) shall be installed on the outside of the preserver and shall be connected with the air compartment by means of a heat sealable connector 3/8-inch (minimum) inside diameter, 9/16-inch outside diameter, which shall in turn be heat sealed to the inside surface of the preserver fabric. The construction of the oral inflation assembly shall be in accordance with Drawing 803-2482705. The location of the oral inflation assembly shall be as shown on Drawing 803-2482705. The valve shall be secured to the connector by a clamp capable of withstanding a load of 50 pounds. The design of the securing device shall be submitted to the command or agency concerned for tentative approval.

3.5 Detail requirements.

3.5.1 Design, manufacture, and assembly. The design, manufacturing, and assembly details of the vest type preservers shall conform to Drawing 803-2482705.

3.5.2 Mechanical inflation system.

3.5.2.1 Inflation assembly. Each life preserver shall be provided with a CO_2 inflation assembly conforming to type III of MIL-I-23145. The assembly shall use CO_2 cylinders conforming to type II of MIL-C-601. Inflation assemblies shall be safety-wired using a length of wire conforming to QQ-W-343 type S, 22 AWG drawn and annealed.

3.5.2.2 Inflation assembly lanyard. The lanyard assembly for the type III inflation assembly shall conform to MIL-I-23145, except that the cord shall be a 1/16-inch diameter commercial braided nylon rope. The discharge levers, rope, and button shall withstand a minimum pull of 100 pounds.

3.5.2.3 Manifold. The inflation assembly shall be securely attached to the preserver by means of a brass manifold with a heat sealable base, which shall in turn be heat sealed to the inside surface of the preserver fabric, so that the inflation assembly projects on the outside. The brass manifold, including threads, shall be coated with electrodeposited nickel conforming to class 1, grade E of QQ-N-290. The manifold and inflation assembly shall be equipped with flats to maintain the inflation assembly in a fixed position. The manifold shall conform to piece number 25 of Drawing 803-2482705 and shall fit the inflation assembly. The manifold shall be located and oriented as shown on Drawing 803-2482705. The manifold shall be provided with a one-way check valve conforming to Drawing 805-1183395. Two synthetic rubber gaskets conforming to MIL-R-6855 shall be provided, one for each side of the inflation assembly attachment. The hexagonal nut shall be nickel-plated brass.

3.5.3 Oral inflation valve and tube. The oral inflation tube shall be in accordance with type II, class 1, style A of MIL-O-81375, except that the tube shall withstand without damage, inflation by means of a blow gun having a

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discharge nozzle not exceeding 1/8 inch diameter, when the gun is connected to a 225 pound per square inch lb/in^2 pressure source.

3.5.4 Adhesive. Any adhesive used in the fabrication of the preserver shall be polychloroprene conforming to class 1, 2, 3, or 4 of MIL-A-5540. The color shall acceptably match the preserver coated cloth.

3.6 Label. Each vest shall have an identification label measuring 1-5/16 inches wide by 2-3/4 inches long. The labels shall conform to type VI, class 1 of DDD-L-20 and shall have the following inscription and size of characters:

Inscription	Size of characters
Life preserver, identification	18-point Gothic bold
Vest type, U.S. Navy, Mark I	18-point Gothic bold
Contract No. xxxx (example)	10-point Gothic medium
Size - Medium (example)	10-point Gothic medium
Manufacturer's name	10-point Gothic medium

3.6.1 On items fabricated with nylon covers (type II), the first line of the identification label shall read as follows:

Inscription	Size of characters
Life preserver	18-point Gothic bold

3.7 Size. Vest type life preservers shall be sized as small, medium, and large (see 1.2.1).

3.8 Color (see 1.2.2). Colors 1 through 7 shall match the standards shown in Standard Color Card of America, 9th edition. Colors 8 and 9 shall match the color chips 12246 and 36081, respectively, shown in FED-STD-595.

3.9 Marking. Identification marking shall be 1/4-inch high letters and numerals made by the silk screen method. In all cases the ink shall be a contrasting color, waterproof, and shall cause no deterioration of any fabric.

3.9.1 Identification. The identification shall consist of the name of the item, serial number, name or trademark of the manufacturer, and the date of manufacture (month and year). The identification shall be located on the inflatable bladder as shown on Drawing 803-2482705.

3.9.2 Serial numbers. The vest and/or bladder shall be identified by numbers which shall be assigned by the manufacturer. (see 6.2.1) Serialization shall be a block of consecutive numbers to cover the entire acquisition document quantity. Method of marking shall be specified in 3.9.

3.10 Workmanship. Workmanship shall be in accordance with high-grade manufacturing practice covering this class of work. Because of the emergency use of this equipment, it is required that all details of workmanship be of the highest quality in order to ensure proper functioning of the life preservers at all times.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality control. To ensure that the final product is basically sound and in strict conformance with this specification, the contractor shall provide for an adequate control of quality of the coated and uncoated fabric, the adhesive, and the attachments during the process of manufacture. The results of pressure, leakage, and buoyancy tests shall be identifiable by listing the assigned serial numbers (see 3.9.2).

4.1.3 Parts and material inspection. The contractor shall be responsible for ensuring that parts and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. The inspection performed on the completed preserver, parts, material, or inflatable bladder shall consist of all tests specified in this specification, including examination (see 4.7 and 4.10), to determine strict conformance to the construction requirements. Samples of all material used in the fabrication of all preservers shall be examined and tested to ascertain their conformance to the applicable specifications or the requirements for the material as specified herein. The test for qualification approval of the vest (cover) only shall consist of the examination applicable to the vest (cover) of 4.7 and the salt spray test of 4.10.11.

4.3.1 Samples for qualification inspection and first article inspection of complete life preserver. The following samples are required for qualification and first article inspection of complete life preserver:

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(a) Two complete life preservers.
(b) Two cured pieces of preserver coating compounds, 6 by 6 inches and 0.07 to 0.08 inch thick.
(c) 2 yards each of all coated and uncoated cloth used in the inflatable bladder and preserver cover. The name of the supplier and his code number shall be furnished along with these samples.
(d) Two coating adhesion samples, 12 by 12 inches, prepared as specified in MIL-C-19002.
(e) Four adhesion samples, 7 by 8 inches, of the coated bladder cloth, prepared with the cement used in the preserver (two of the samples prepared coated face to coated face and two prepared uncoated face to coated face). If heat sealing is used in attachment of parts to the bladder, two heat sealed samples, 7 by 8 inches, prepared coated face to coated face, shall also be furnished.
(f) Four seam adhesion samples of the bladder cloth, 12 by 12 inches, with a 1/8-inch wide seam running along the centerline of the entire width of the sample.
(g) 5 yards each of all webbing and tape used on the preserver cover.
(h) One cone (500 yards) of all thread used in the preserver.
(i) Four sets of all hardware used in the preserver, including the oral and CO_2 inflation systems.
(j) 5 yards of safety wire.

4.3.1.1 Samples for qualification inspection and first article inspection of inflatable bladder. The following samples are required for qualification or first article inspection of inflatable bladder as required:

(a) Two inflatable bladders.
(b) Two cured pieces of preserver coating compounds, 6 by 6 inches and 0.07 to 0.08 inch thick.
(c) 2 yards each of all coated and uncoated cloth used in the inflatable bladder. The name and code number of the supplier shall be furnished along with these samples.
(d) Two coating adhesion samples, 12 by 12 inches, prepared as specified in MIL-C-19002.
(e) Four adhesion samples of the coated bladder cloth, 7 by 8 inches, prepared with the cement used in the preserver; (two of the samples prepared coated face to coated face and two prepared uncoated face to coated face). If heat sealing is used in attachment of parts to the bladder, two heat sealed samples, 7 by 8 inches, prepared coated face to coated face, shall also be furnished.
(f) Four seam adhesion samples of the bladder cloth, 12 by 12 inches, with a 1/8-inch wide seam running along the centerline of the entire width of the sample.
(g) One cone (500 yards) of all thread used in the bladder.
(h) Four sets of all hardware used in the bladder, including the oral inflation systems.

4.3.1.2 Samples for qualification inspection and first article inspection of vests (covers). The following samples are required for qualification or first article inspection on vests (covers):

(a) Two complete vests (covers)

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(b) 2 yards each of all coated and uncoated cloth used in the vest (cover). The name of the supplier and his code number for these items shall be furnished along with the samples.
(c) 5 yards each of all webbing and tape used on the vest (cover).
(d) One cone (500 yards) of all thread used in the vest (cover).
(e) Four sets of all hardware used in the vest (cover).

4.3.2 Qualification test samples shall be plainly identified by securely attached durable tags marked for forwarding to a testing laboratory acceptable to the command or agency concerned with the following information:

"(a) Samples for qualification tests.
(b) (Name of product together with complete identifying data such as manufacturer's designation or number.)
(c) (Name of manufacturer.)
(d) Submitted by (name) (date) for qualification inspection in accordance with the requirements of specification (no.) under authorization (reference authorizing letter)."

4.4 First article inspection. First article inspection shall be performed on the sample submitted as specified in 3.2. First article inspection of complete life preservers or inflatable bladders shall consist of the examination and all the tests of this specification (see 4.7 and 4.10). First article inspection of vest (cover) shall consist of the examination applicable to the vest (cover) of 4.7 and the salt spray test of 4.10.11.

4.4.1 Government inspection. When the contract or order specifies that first article inspection be performed by the Government (see 6.2), the contractor shall submit the sample and a set of prints of the manufacturing drawings to the laboratory specified (see 6.2) for the performance of first article inspection.

4.4.2 Contractor inspection. When the first article inspection is performed by the contractor, the inspection shall be witnessed (see 6.2.1) and the test results verified by the inspector.

4.4.3 First article inspection report. For each acquisition, the contractor (or the Government laboratory, see 4.4.1) shall submit to the contracting activity, for approval, a complete report of all the first article tests conducted (see 6.2). One copy of this report shall be forwarded to NAVSEA and the cognizant Defense Contract Administration Services Management Area (DCASMA). Reports shall show quantitative results of all tests and analyses required by this specification. Test reports shall be in duplicate.

4.5 Quality conformance inspection.

4.5.1 Each unit tests. The following tests shall be conducted as a part of the quality control (see 4.1.2) on each vest type preserver. When specified (see 6.2), tests shall be witnessed by the Government inspector.
Tests shall be performed in the order listed.

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Test	Quantity	Paragraph
Operation (CO_2)	Each	4.10.1
Operation of oral valve	Each	4.10.2
Pressure (5 lb/in^2)	Each	4.10.3
Leakage (1 lb/in^2)	Each	4.10.4
Operation of relief valve	Each	4.10.5

Preservers not conforming to the tests shall be reworked prior to submission to the sampling inspection (see 4.5.3), but under no circumstances shall an air retaining seam be reworked more than once.

4.5.1.1 The following additional tests shall be conducted.

Test	Sampling plan	Paragraph
Leakage and operation (inflation assembly)	Each	4.10.7
Oral inflation tube (porosity)	Each	4.10.8.1
Oral inflation tube (tensile)	6 samples per 1000	4.10.8.2
Dimensional examination of inflation assembly	Each	4.10.9
Oral inflation tube (valve holding strength)	6 samples per 1000	4.10.12

4.5.2 Adhesion test samples. At the beginning of each day's production, one bladder shall be sealed and tested in accordance with the applicable test of 4.10.10. If the sample passes the test, production may commence; if a failure occurs, corrective action shall be taken to determine and correct the cause. Upon commencing production, one bladder from each 100 production bladders sealed shall be tested in accordance with the applicable test of 4.10.10. If, during the cycle, bladders of fabric from a different fabric coater are to be sealed, a fresh sample shall be tested and the cycle count returned to 0. If failures occur during any sample tests, production shall be halted and corrective action shall be taken to determine and correct the cause.

4.5.2.1 Cemented test samples. For the test of cemented samples of 4.10.10.1, the dimensions of the test specimen shall be in accordance with the requirements for strip specimens as specified in method 8011 of FED-STD-601.

4.5.3 Sampling inspection.

4.5.3.1 Examination. The examination of 4.7 shall be conducted on samples selected in accordance with 4.6.2. Any sample which contains one or more defects shall not be offered for delivery, and if the number of nonconforming items exceeds the acceptance number for that sample, the entire lot shall be rejected.

4.5.3.2 Tests. The tests of 4.10.1 through 4.10.6 shall be conducted on samples selected in accordance with 4.6.3 which have passed the examination (see 4.7) and the tests of 4.5.1, 4.5.1.1, and 4.5.2. When specified (see 6.2), tests shall be witnessed by the Government inspector. Any item in the sample containing one or more defects shall not be offered for delivery. If the number

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of nonconforming items in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5.3.2.1 Tear strengths. The minimum tear strengths (see 3.3.1.2) shall be 7 pounds for both warp and filling. Maximum allowable shrinkage shall be 2 percent for both warp and filling. Test methods for these physical requirements shall be in accordance with FED-STD-191.

4.5.3.2.2 Buoyancy test. The buoyancy test (see 4.10.6) shall be conducted when the CO_2 cylinders are expended during the operation test (see 4.10.1).

4.5.4 Comparison tests. When specified (see 6.2.1), NAVSEA may conduct, at a Government laboratory, comparison tests on random samples of preservers produced under any contract. Comparison tests may include any or all of the tests of this specification. Acceptance and shipment of preservers shall not be delayed pending completion of comparison tests. If, however, any comparison test sample is rejected for noncompliance with the requirements of this specification, acceptance and shipment of any further preservers shall be withheld until the contractor can demonstrate that he has corrected the conditions which led to the rejection of the comparison test sample.

4.6 Sampling for quality conformance inspection.

4.6.1 Lot. A lot shall consist of all preservers of the same size and color submitted for inspection at one time. Each lot shall be tested in accordance with 4.6.3.

4.6.2 Sampling for examination (major and minor defects). A random sample of life preservers shall be selected from each lot in accordance with inspection level I of MIL-STD-105 for the examination for major and minor defects in 4.7. The Acceptable Quality Level (AQL) shall be 4.0 for major defects and 10.0 for minor defects per 100 units.

4.6.3 Sampling for tests. A random sample of preservers shall be selected from each lot in accordance with the following inspection levels as specified in MIL-STD-105 for tests specified in 4.5.3.

Tests	Inspection level
Operation (CO_2 see 4.10.1)	I
Operation of oral valve (see 4.10.2)	I
Pressure (5 lb/in^2) (see 4.10.3)	I
Leakage (1 lb/in^2) (see 4.10.4)	I
Buoyancy (see 4.10.6)	S-3
Preserver cover (see 3.4)	S-1

The acceptance number shall be zero for all tests. Rejected preservers shall not be resubmitted.

4.7 Examination. Life preservers shall be thoroughly examined to determine conformance with this specification and the applicable drawings.

4.7.1 For critical defects. The life preservers shall be examined 100 percent for critical defects as defined in table III. The acceptance number shall be zero.

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TABLE III. Classification of defects.

Categories Critical:	Defects
1.	Buoyancy chamber material defective; evidence of splits, tears, patched holes, and fraying.
2.	Buoyancy chamber seams defective, incomplete, or not as specified; evidence of bubbles, incomplete cementing, or heat sealing; seams in buoyancy chamber not the specified type; width of seams less than the required minimum.
3.	Manifold or oral inflation tube defective, not secure, securing patches not as required, or diameter of base not as specified.
4.	Oral inflation valve defective or not secure in tube; tube evidences cracks, splits, bubbles, bulges, pin holes, or other deleterious effects.
5.	Inflation assembly defective, not securely attached, or not as specified; lanyard not securely attached to actuating lever; level holding pin not staked in place.
6.	Failure of specified gauge to seat firmly against seating surface of inflation assembly; O-rings for cylinder and piercing pin missing, defective, or not as required (inspect prior to assembly).
7.	Gaskets (each side of inflation assembly attachment) missing, defective, or not as required.
8.	Snap fasteners not properly installed, of wrong type, or incomplete.
9.	Parts damaged, incorrect, missing, misfitting, or faulty, impairing the function of the preserver as an item of lifesaving equipment.
Major:	
101	Loop for securing inflation tube defective or not securely attached as specified.
102	Inflation assembly not secured as required. Inflator cover not secure or snap fasteners missing, defective, or not secured to cover as required.
103	Webbing evidences tears, splits, fraying, or weak spots.
104	Stitching defective, incomplete, or not as specified, unevenly guaged, or varies more than 1/16 inch when range not specified.
105	Hardware missing, defective, or not secure; evidence of cracks, deep pits, thin spots, or sharp surfaces.
106	Inflation assembly actuating level not safety-wired as specified.
107	Metal parts (webbing hardware, holder, lever, or manifold) not finished as specified; finish defective or incomplete.
108	Date of manufacture marking missing, illegible, or incorrect.
109	Set screws provided in inflation assembly for securing CO_2 cylinders missing or not properly installed.
110	Evidence of reworking air retaining seams more than the allowable number of times.
111	Inflation assembly threads not greased as required after assembly on preserver.
112	Bartacks missing, loose, or misplaced, not serving intended purpose.
113	Evidence of needle chew which may develop into a hole.
114	Evidence of hole, cut, tear, mend, or burn.

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TABLE III. Classification of defects. - Continued

Categories Critical:	Defects
115	Any part of garment caught in any unrelated operation or stitching.
116	Stitch tension incorrect resulting in: a loose seam, loosely exposed bobbin thread or top thread, or puckering which is evidence of tight tension.
117	Weakening defect such as a smash, multiple float, or loose slub that may develop a hole.
118	Tape fasteners missing, defective, or not securely attached as specified.
119	Reflective tape missing, loose, misplaced, or not of the size specified.
201	Length of lanyard not as specified.
202	Color of cloth, webbing, thread, and adhesive not as specified.
203	Marking, manufacturer's name or trademark and identification missing, incorrect, illegible, not permanent, or not located as specified.
Minor:	
204	Completed preserver not clean; one of more spots or stains which are not removable and are clearly noticeable.
205	Thread ends not trimmed, loose threads not removed from major portion of vest.
206	Seams, twisted, puckered, or pleated, affecting appearance.
207	Binding not securely caught on fabric, easy to pull away from fabric.
208	Binding tight, causing excessive puckers or pleats, affecting appearance.
209	Packaging and packing not as specified.
210	Adhesive on fabric surface around accessory patches below minimum length of 1/8 inch or in excess of 3/8 inch.
211	Preservers improperly folded.

4.7.2 For major and minor defects. Sample preservers selected in accordance with 4.6.2 shall be examined for major and minor defects as defined in table III.

4.7.3 Resubmitted lots. Resubmitted lots allowed in connection with major and minor defects found in the examination may be resubmitted for examination as specified in 4.6.2. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be selected with the acceptance number remaining zero. The resubmitted lot shall be examined by the contractor and witnessed by the Government inspector when specified (see 6.2.1) using tightened examination. Rejected lots may only be resubmitted for acceptance after complete screening and correction for noted defects.

4.8 Test conditions.

4.8.1 Atmospheric conditions. Unless otherwise specified in the contract or order (see 6.2.1), all tests required by this specification shall be

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conducted at prevailing atmospheric pressure, a temperature of 77 ± 18 degrees Fahrenheit (F), and a relative humidity of 80 percent or less. If the final values for ambient temperature or barometric pressure in the 24 hour air pressure tests are different from the initial values, the following corrections shall be made to the final pressure readings in lb/in^2:

(a) Temperature corrections. For each degree rise in temperature, 0.031 lb/in2 shall be subtracted from the final pressure reading. For each degree drop in temperature, 0.031 lb/in^2 shall be added to the final pressure reading.
(b) Barometric pressure correction. For each 0.1 inch of Mercury (Hg) rise in barometric pressure, 0.049 lb/in2 shall be added to the final temperature-corrected pressure reading. For each 0.1 inch of Hg drop in barometric pressure, 0.049 lb/in^2 shall be subtracted from the final temperature-corrected pressure reading.
(c) Pressure readings. Pressure shall be measured by means of a Hg manometer and then converted to lb/in^2 by multiplying inches of Hg by 0.49.

4.8.2 CO_2 cylinders for test purposes. CO_2 cylinders used in conducting the tests specified in 4.10.1 through 4.10.6 shall be in accordance with type II of MIL-C-601. The cylinders shall be marked with a permanent type ink as follows: "For Test Purposes Only." The inclusion of a nonapplicable contract number or the omission of the contract number is not acceptable.

4.9 Possible test failures. Possible test failures shall be determined as follows:

(a) Operation of assembled life preserver:

(1) Failure of life preserver to inflate when lanyard is pulled; evidence of leaks through fabric, seams, inflation assembly or piercing pin, O-rings, or manifold check valve after inflation.
(2) Life preserver cannot be readily inflated through the oral inflation valve, or there is evidence of leakage back through valve after inflation.
(3) Evidence of material failure or parting of seams when the preserver is subjected to the specified pressure test.

(b) Leakage: The pressure within the life preserver is less than the minimum permissible after it has been inflated and then allowed to stand for the required length of time.
(c) Buoyancy: Failure of the inflated life preserver to support, without sinking, the specified weight.
(d) Pull: Discharge lever, rope, and button fail to withstand the required minimum pull.

4.10 Tests.

4.10.1 Operation (CO_2). The operation of the preserver shall be tested by releasing the CO_2 cylinders into the life preserver in the normal manner. The life preserver shall fully inflate in not more than 5 seconds.

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4.10.2 Oral valve operation. The preserver shall be inflated through the oral valve by means of a blow gun having a discharge nozzle not exceeding 1/8-inch diameter when the gun is connected to a 90 ± 5 lb/in^2 pressure source. The preserver shall be inflated until the relief valve opens. The oral valve shall work without difficulty, shall shut off securely, and shall not leak underwater with the knurled nut in the open position.

4.10.3 Pressure. The preserver shall be inflated with air through the oral inflation valve to a pressure of 5 lb/in^2 and shall be allowed to stand for a minimum of 5 minutes. The pressure drop shall be no greater than 0.4 lb/in^2. The preserver shall be examined while held under pressure. There shall be no evidence of failure of the material in any respect. All seams and attachments shall remain intact and shall show no signs of separation.

4.10.4 Leakage. The life preserver shall be inflated with air through the oral inflation valve to a pressure of 1 lb/in^2 and allowed to stand for a minimum of 15 minutes. The pressure shall then be checked and readjusted, if necessary, to the original pressure of 1 lb/in^2. The temperature and barometric pressure shall be recorded at this time. At the end of a minimum of 24 hours after the readjustment period, the pressure shall be measured and corrected for any changes in temperature or barometric pressure (see 4.8.1). The corrected pressure shall be not less than 0.75 lb/in^2. The oral inflation system shall be examined when the preserver is being deflated to determine that the valve works without difficulty and shuts off securely.

4.10.5 Buoyancy. Immediately upon completion of the operational (CO_2) test (see 4.10.1), the preserver shall be attached to a weight weighing 29 pounds when submerged in fresh, clean water. The preserver shall support this 29-pound weight without sinking.

4.10.7 Operation (inflation assembly). CO_2 inflation assemblies shall be subjected to all the tests specified in section 4 of MIL-I-23145 to determine conformance with the requirements specified therein.

4.10.8 Oral inflation tube. Before the oral inflation valve is installed in the oral inflation tube, the tests specified in 4.10.8.1 and 4.10.8.2 shall be conducted.

4.10.8.1 Porosity. The oral inflation tube shall be subjected to an air pressure of 25 lb/in^2 for a minimum of 10 seconds while submerged under water. There shall be no evidence of air leakage due to porosity. There shall be no evidence of weakening of the tube as revealed by excessive deformation in any portion of the tube.

4.10.8.2 Tensile. The oral inflation tube base shall be folded longitudinally so that the rounded edges coincide, and then placed in the lower jaws of a test machine. The tube portion shall be placed in the upper jaws. The rate of load application shall be 12 ± 1/2 inches per minute. The tensile strength shall be not less than 75 pounds.

4.10.9 Dimensional examination (gauge) of inflation assembly. Inflation assembly (complete) shall be dimensionally checked with a cylinder gauge as specified in MIL-I-23145. Failure of the plug gauge to seat firmly against seating surface (where the puncturable end of the cylinder seats within the inflation assembly) shall be cause for rejection.

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4.10.10 Adhesion.

4.10.10.1 Cemented samples. The test for adhesion of cemented samples shall be conducted in accordance with method 8011 specified in FED-STD-601. The adhesive strength of the cemented samples shall be not less than 7.5 pounds per inch of width for samples joining the coated sides of the finished fabric and not less than 5.0 pounds per inch of width for seams joining the coated side to the uncoated side of the finished fabric.

4.10.10.2 Heat sealed samples. The sample heat sealed bladder seams shall be subjected to the adhesion test specified in AMS 3272 Type I.

4.10.11 Salt spray. A 100-hour salt spray test shall be conducted on the oral valve, CO_2 inflation device, and all metal parts in accordance with ASTM B 117. There shall be no evidence of corrosion which will adversely affect the operation of the equipment.

4.10.12 Oral inflation valve holding strength. This test shall be conducted after the oral inflation valve is clamped in the oral inflation tube. The oral inflation tube base shall be folded longitudinally so that the folded edges coincide, and then placed in the lower jaws of a testing machine. The oral inflation valve shall be placed in the upper jaws. The rate of load application shall be 12 ± 1/2 inches per minute. The load to separate the oral inflation valve from the oral inflation tube shall be not less than 50 pounds.

4.11 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging. Packaging shall be level A or C (see 6.2).

5.1.1 Level A. Each preserver, complete with charged CO_2 cylinders (see 4.8.2), shall be folded as follows:

(a) The inflatable bladder shall be inserted into the vest portion so that all parts are properly oriented and the bladder has no twists or lumps.
(b) The CO_2 inflation system shall be properly installed.
(c) The inflatable bladder shall be properly evacuated through the oral inflation system and the oral inflation valve locked by means of the knurled ring.
(d) All fasteners shall be closed and the garment laid flat, face up.
(e) One side of the vest shall be folded in so that the width of the garment is approximately 11-1/2 inches.
(f) The top of the garment shall be folded down so that the folded garment measures approximately 11-1/2 by 15-1/4 inches.

5.1.2 Level C. The vests shall be packaged in a manner that will afford protection against deterioration and mechanical or physical damage during

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shipment from supply source to the first receiving activity. The contractor may use his own practice providing it fulfills these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twelve boxes of vests, packed as specified in 5.2.2, shall be overpacked six in length, one in width, and two in depth, in a fiberboard box that conforms to class weather-resistant, size number 4 of MIL-B-17757. Closure and reinforcing requirements for the boxes shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty vests, of one size and color only, folded as specified in 5.1.1, shall be tiered flat in a shipping container, with each vest reversed, and the larger dimension of the folded vests parallel to the side panels of the box. The top and bottom of each pack of vests shall be completely covered with a sheet of kraft paper of not less than 30 pounds basis weight conforming to grade B of UU-P-268. The fiberboard shipping container shall conform to class domestic, size number 1 of MIL-B-17757. Closure requirements for the boxes shall be as specified in the appendix to the box specification.

5.2.3 Level C. Vests shall be packed in a manner which will ensure acceptance and safe delivery at destination. Shipping container shall conform to carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. Shipping containers shall be marked in accordance with MIL-STD-129. When level A packing is specified, the fiberboard box (see 5.2.2) shall be marked as specified for interior packages, and the overpack box (see 5.2.1) shall be marked as specified for shipping containers.

5.3.1 Nomenclature. The following nomenclature shall be marked on the shipping containers: Life Preserver, Vest Type, U.S. Navy, Mark I. Size and color shall also be indicated.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The preservers covered by this specification are intended for wear by flight deck personnel for identification and survival purposes (type I)(colors 1 through 7), and as an inflatable life preserver (type II)(colors 8 and 9) where their use is authorized.

1	White (type I)	-	Phone talkers and medical corpsmen
2	Red	-	Ordnance men
3	Green	-	Catapult and arresting gear personnel
4	Blue	-	Plane pushers
5	Brown	-	Plane captains and mechanics
6	Yellow	-	Flight deck officers and plane directors
7	Purple	-	Fuel handlers
8	Safety orange	-	Life preserver
9	Gray	-	Life preserver

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6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
(c) Color, size, and type required (see 1.2.1, 1.2.2, and 1.2.3).
(d) When first article inspection is required (see 3.2).
(e) Color of cotton and nylon uncoated cloth (see 3.3.1.2.1 and 3.3.1.2.2).
(f) When first article inspection is to be performed by the Government (see 4.4.1).
(g) Laboratory for first article inspection when performed by the Government (see 4.4.1).
(h) When first article tests done by the contractor are to be witnessed by the Government inspector (see 4.4.2).
(i) When quality conformance tests are to be witnessed by the Government inspector (see 4.5.1 and 4.5.3.2).
(j) When comparison tests are to be conducted (see 4.5.4).
(k) When examination of resubmitted lots is to be witnessed by the Government inspector (see 4.7.3).
(l) When atmospheric conditions of tests are to be other than as specified (see (4.8.1).
(m) Level of packaging and packing required (see 5.1 and 5.2).
(n) Serial numbers to be assigned by manufacturer (see 3.9.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure corrent contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Referenced Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.3	DI-T-4902	First Article Inspection Report	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for

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examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24247 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6". (See 6.5.1)

6.4.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Buoyancy
Carbon dioxide
CO_2 cylinders
Inflatable bladders
Life preservers
Oral inflation
Vest

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4220-N320)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-24247C(SH)	Life Preserver, Vest Type U.S. Navy, Mark I

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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